

Problem statement on Langflow

Problem Statement No.1: Nutrition Agent

The Challenge

Good nutrition is the foundation of health, but individuals often struggle with **diet planning, portion control, and tracking nutritional intake**. Information about healthy diets, disease-specific nutrition, and food composition is scattered across unreliable sources. Without **personalized, real-time guidance**, many fail to maintain balanced diets or make informed food choices.

The Objective

Build an **AI-powered Nutrition Agent** that delivers **personalized dietary insights and guidance** for individuals, families, and health professionals. The solution should:

1. **Personalized Diet Plans** :- Recommend meals based on age, health conditions, allergies, cultural preferences, and fitness goals.
2. **Food Data RAG Retrieval** :- Fetch and summarize nutritional values (calories, macros, vitamins) from reliable sources.
3. **Diet Tracking & Feedback** :- Allow users to log meals via text, images, or voice, and provide instant nutritional analysis.
4. **Preventive Health Focus** :- Suggest diets for chronic disease management (e.g., diabetes-friendly or heart-healthy plans).
5. **Multi-Agent System** :-
 - Nutrition Knowledge Agent (fetches nutritional data)
 - Diet Recommendation Agent (personalized meal plans)
 - Health Advisory Agent (preventive health suggestions)
 - Food Log & Feedback Agent (meal tracking + analysis)
6. **Visualization Dashboard** :- Show daily/weekly nutrient intake, deficiencies, and progress toward health goals.

Tech Stack (Mandatory)

Use **any one** of the following approaches:

- LangFlow + IBM watsonx.ai + Granite Models
- IBM Cloud Agentic Lab with IBM Granite Models
- Replicate with LangChain + RAG + IBM Granite Models